Unit-2

PROBABILITY

In our normal conversation, we make statements like "I am likely to be out of station", "I have a good chance of being selected to the job", etc. In such statements, we are not sure of the outcome. But we intend to estimate the chances of ~~being~~ our statement being true. Probability gives an insight for such things in a mathematical way.

① Exhaustive events: An event consisting of all the various possibilities is called the exhaustive event.

② Mutually exclusive events: Two or more events are said to mutually exclusive if the happening of one event prevents the simultaneous happening of the others.

③ Independent events: Two or more events are said to independent if the happening or the non-happening of one event does not prevent the ~~,~~ the others.

If the outcome of a trail consists 'n' exhaustive, mutually exclusive, equally possible cases of which m of them are favourable to event E, then probability of happening of event E is, P(E) is equal to $\frac{m}{n}$.

i.e, $P(E) = \frac{\text{no. of favorable cases}}{\text{no. of possible cases}}$

$$P(E) = \frac{m}{n}$$

The probability can almost can be equal to 1, b/z the no. of favourable cases & the no. of possible cases can almost coincide with each other.

Since m cases are favourable to the event E, it follows that n-m cases are not favorrable to event E. The set of unfavourable events is denoted by $\bar{E}$. The probability of the non-happening of the event, $P(\bar{E}) = \frac{n-m}{n} = \frac{n}{n} - \frac{m}{n} = 1 - \frac{m}{n} = 1 - P(E)$

i.e, $P(E) + P(\bar{E}) = 1 \quad (p+q=1)$

p is also referred as the probability of success and
q " " " failure, their sum = 1.

Eg:- 1) The probability of getting head in tossing a coin $= \frac{1}{2}$

The possible outcomes are head and tail.

The no. of possible cases $(n) = 2$

The no. of favourable cases $(m) = 1$

$\therefore P(E) = \frac{1}{2}$

2) The probability of getting i) king ii) king or queen, when a card is replaced, when a card is drawn from the pack of 52 cards.

i) Total possible cases $(n) = 52$

No. of favourable cases $(m) = 4$

$\therefore$ Probability of getting king $= \frac{m}{n} = \frac{4}{52} = \frac{1}{13}$.

ii) The no. of favourable cases $(m) = 4+4 = 8$

$\therefore P(E) = \frac{8}{52} = \frac{2}{13}$

3) The probability of getting

i) a no. greater than two
ii) an odd no.
iii) an even no., when a die is thrown

i) E : no. > 2

$m = 4 \quad n = 6 \quad \therefore P(E) = \frac{4}{6} = \frac{2}{3} = 0.666$

ii) E : odd no. $(1, 3, 5)$

$m = 3, \; n = 6 \quad P(E) = \frac{3}{6} = \frac{1}{2} = 0.5$

iii) E : even no. $(2, 4, 6)$

$m = 3, \; n = 6 \quad P(E) = \frac{3}{6} = \frac{1}{2} = 0.5$

4) When 2 dice are thrown simultaneously, what is the prob. of

i) getting a total more than 10.
ii) " " less than 10.
iii) " " equal to 10.

ii) E: getting total more than 10.

E: (6,5), (5,6), (6,6)

$m = 3$, $n = 36$ $\therefore P(E) = \frac{3}{36} = \frac{1}{12}$

i) E': getting less than 10. 10 = (4,6), (6,4), (5,5)

$P(E') = 1 - P(E) = 1 - \frac{30}{36} = \frac{11}{12} \cdot \frac{10}{12} = \frac{5}{6}$

iii) E: getting total equal to 10

E: (4,6), (6,4), (5,5)

$\therefore P(E) = \frac{3}{36} = \frac{1}{12}$

5) What is the prob. that a leap year will have 53 sundays?

soln: In a leap year, there are 366 days. When it comprises of 52 weeks & 2 days. These 2 days can be (S, M), (M, T), (T, W), (W, Th), (Th, F), (F, Sat), (Sat, S).

The no. of possible outcomes (n) = 7 and no. of favourable cases (m) = 2

$\therefore$ Prob. of $= \frac{m}{n} = \frac{2}{7}$

Addition theorem

The probability of the happening of one or the other mutually exclusive events is equal to the sum of the probabilities of the 2 events.

i.e, If A and B are two mutually exclusive events, then

$$P(A \text{ or } B) = P(A) + P(B)$$

Let the total number of possible cases in the trials be n. Out of these let m_1 cases be favourable to A and m_2 cases to B.

Hence the no. of cases favourable to either A or B is $(m_1 + m_2)$.

$$\therefore P(A \text{ or } B) = \frac{(m_1 + m_2)}{n} \quad \text{---} \quad (1)$$

Clearly $P(A) = \frac{m_1}{n}$ and $P(B) = \frac{m_2}{n}$

$$\therefore P(A) + P(B) = \frac{m_1}{n} + \frac{m_2}{n} = \frac{m_1 + m_2}{n} \quad \text{---} \quad (2)$$

Thus from (1) & (2) $P(A \text{ or } B) = P(A) + P(B)$

Product theorem:

If a compound event is made up of a number of independent events, the prob of happening of the compound event is equal to the product of the probabilities of the independent events.

i.e., Since $P(A \text{ and } B) = P(A) \cdot P(B)$.

Proof: Since A and B are independent events, Let the event A occurs in a_1 ways and fails in b_1 ways.

Let the event B occurs in a_2 ways and fails in b_2 ways.

$\therefore P(A) = \frac{a_1}{a_1+b_1}$ & $P(B) = \frac{a_2}{a_2+b_2}$

There are 4 possibilities for the compound event.

1. A occurs & B occurs in $a_1 a_2$ ways
2. A occurs & B does not occur in $a_1 b_2$ ways
3. A does not occur & B occurs in $b_1 a_2$ ways
4. A does not occur & B does not occur in $b_1 b_2$ ways.

$\therefore$ The no. of possible ways for the compound event is

$$a_1a_2 + a_1b_2 + b_1a_2 + b_1b_2$$
$$= a_1(a_2+b_2) + b_1(a_2+b_2)$$
$$= (a_1+b_1)(a_2+b_2)$$

The no of ways favourable to the compound event is $a_1 a_2$ And hence the prob of happening of the comp. event is

$$P(A \text{ and } B) = \frac{a_1 a_2}{(a_1+b_1)(a_2+b_2)} \quad \text{---} (1)$$

$$= \frac{a_1}{a_1+b_1} \cdot \frac{a_2}{a_2+b_2}$$

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Note: (1) If $A_1, A_2, A_3, A_4 \ldots A_n$ are mutually exclusive events, then

$P(A_1 \text{ or } A_2 \text{ or } A_3 \text{ or } \ldots A_n) = P(A_1) + P(A_2) + P(A_3) + \ldots P(A_n)$

② If $A_1, A_2, A_3, \cdots A_n$ are independent events, then

$P(A_1 \text{ and } A_2 \text{ and } A_3 \text{ and } \cdots A_n) = P(A_1) \cdot P(A_2) \cdot P(A_3) \cdots P(A_n)$

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1) A box contains 3 white, 5 blacks and 6 red balls. If a ball is drawn at random, what is the prob that it is either red or white?

soln:- total no. of balls in the box $(n) = 14$

Probability of getting red ball = $P(\text{getting red}) = \frac{6}{14}$

$P(\text{white ball}) = \frac{3}{14}$

$\therefore P(\text{red or white}) = \frac{6}{14} + \frac{3}{14} = \frac{9}{14}$

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2) Two cards are drawn in succession from a pack of 52 cards. find the prob that the first is king and the second is queen, if the first card is i) replaced ii) not replaced.

$P(\text{king}) = \frac{4}{52}$

(replaced)

i) $P(\text{queen}) = \frac{4}{52}$

$P(K \text{ and } Q) = \frac{4}{52_{13}} \cdot \frac{4}{52_{13}}$

$= \frac{1}{169}$

(ii) when 1st card is not replaced

$P(Q) = \frac{4}{51}$

$P(K \cap Q) = P(K) \cdot P(Q)$

$= \frac{4}{52_{13}} \cdot \frac{4}{51}$

$= \frac{4}{663}$

$51 \times 13 = 663$

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3) The prob that the person A solves the problem is 1/3, that of B is 1/2 and that of C is 3/5. If a problem is assigned simultaneously to all of them, what is the prob that the problem is solved?

$P(A) = \frac{1}{3}$ $P(B) = \frac{1}{2}$ $P(C) = \frac{3}{5}$

$P(\bar{A}) = \frac{2}{3}$ $P(\bar{B}) = \frac{1}{2}$ $P(\bar{C}) = \frac{2}{5}$

$P(\text{problem is solved}) = 1 - P(\text{problem is not solved by anyone})$

$= 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$

$= 1 - \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{2}{5}$

$= 1 - \frac{2}{15}$

$= 13/15$ //

*It should be noticed that even if anyone of them solves the problem, it is presumed that the problem is solved.

1) There are 10 students of which 3 are graduates. If a committee of 5 is to be formed, what is the probability that there are
i) only 2 graduates ii) atleast 2 graduates.

Three graduates implies the other seven are non-graduates.
Since a committee of 5 is to be formed, there are $^{10}C_5$ ways of selecting 5 students out of 10.

i) only 2 graduates.

2 graduates implies other 3 are non-graduates.

2 graduates out of 3 can be selected in $^3C_2$ ways.

Any 3 non-graduates out of 7 can be selected in $^7C_3$ ways.

Thus the prob of having only 2 graduates (is equal to) $= \frac{^3C_2 \cdot {}^7C_3}{^{10}C_5}$

$= \frac{3 \times \cancel{35}^{5}}{\cancel{9} \times \cancel{7} \times \cancel{4}} = \frac{15}{36} = \frac{5}{12} = 0.416$

$\frac{\cancel{10}^{2} \times 9 \times \cancel{8} \times \cancel{7} \times \cancel{6}^{2}}{\cancel{4} \times \cancel{3} \times \cancel{2}} \quad \frac{7 \times \cancel{6} \times 5}{\cancel{3} \times \cancel{2}}$

ii) atleast 2 graduates.

Atleast 2 graduates $\Rightarrow$ 2 cases
→ 2 graduates + 3 non-graduates or 3-graduates and 2-non-graduates

The prob of having 3-graduates & 2 non-graduates $= \frac{^3C_3 \cdot {}^7C_2}{^{10}C_5}$

$= \frac{1 \times \frac{\cancel{7} \times \cancel{6}^{3}}{\cancel{2} \times 1}}{\underset{3}{\cancel{9}} \times \cancel{7} \times \cancel{4}} = \frac{1}{12}$

∴ Total prob of committee having atleast 2 graduates $= \frac{5}{12} + \frac{1}{12} = \frac{6}{12} = \frac{1}{2} = 0.5$

2) 3 student X, Y, Z write an examination, their chances of passing are 1/2, 1/3 and 1/4 respectively. Find the prob that i) all of them pass ii) atleast one of them passes iii) atleast two of them pass.

Soln: Let A, B and C be the events that students X, Y and Z resp pass the examination.

$P(A) = \frac{1}{2} \Rightarrow P(\bar{A}) = \frac{1}{2}$

$P(B) = \frac{1}{3} \Rightarrow P(\bar{B}) = \frac{2}{3}$

$P(C) = \frac{1}{4} \Rightarrow P(\bar{C}) = 3/4$

clearly A, B and C are independent events.

i) Prob that all of them pass is equal to $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

$= \frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}$

$= \frac{1}{24}$

ii) Prob that atleast one of them pass = 1 − (Prob that all fail)

$= 1 - P(\bar{A}) \cdot P(\bar{B}) P(\bar{C})$

$= 1 - \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) = 1 - \frac{1}{4} = \frac{3}{4}$

iii) Prob that atleast 2 of them pass $= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) + P(A \cap B \cap C)$

$= P(A) \cdot P(B) P(\bar{C}) + P(A) P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C) + P(A) \cdot P(B) \cdot P(C)$

$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4}$

$= \frac{1}{8} + \frac{1}{12} + \frac{1}{24} + \frac{1}{24} = \frac{1}{4}$

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3) A husband and wife ~~apply~~ appear for 2 vacancies of a post. Prob of husband's selection is 1/7 and that of y is 1/5. What is the prob that

i) both of these will be selected

ii) only 1 of them is selected iii) neither is selected.

Let H and W be the events of selection of husband and wife.

$P(H) = \frac{1}{7}$ $P(W) = \frac{1}{5}$

$P(\bar{H}) = \frac{6}{7}$ $P(\bar{W}) = \frac{4}{5}$

H & W are independent events.

i) $P(H \cap W) = P(H) \cdot P(W) = \frac{1}{7} \times \frac{1}{5} = \frac{1}{35}$

ii) $P(\text{only one}) = P(H \cap \bar{W}) + P(W \cap \bar{H})$

$= \frac{1}{7} \times \frac{4}{5} + \frac{1}{5} \times \frac{6}{7} = \frac{4}{35} + \frac{6}{35} = \frac{10}{35} = \frac{5}{7}$

iii) $P(\text{neither}) = P(\bar{H} \cap \bar{W})$

$= \frac{6}{7} \times \frac{4}{5} = \frac{24}{35}$

4) 5 balls are drawn at random from the bag of 6 white and 4 black balls. what is the chance that 3 of them are white and 2 are black?

Since 5 balls are drawn, there are $^{10}C_5$ ways of seleching 5 out of 10 balls

3 white balls can be selected in $^6C_3$ ways and

2 black balls, , , $^4C_2$ ways.

The required prob $= \frac{^6C_3 \cdot {}^4C_2}{^{10}C_5} = \frac{\frac{6\times5\times4}{3\times2} \times \frac{4\times3}{2}}{\frac{10\times9\times8\times7\times6}{5\times4\times3\times2}} = \frac{20\times6}{252} = 0.476$

$= \frac{10}{21}$

5) A bag contains 4 white and 2 black balls. Another bag contains 3 white and 5 black balls. If a ball is drawn from each, find the prob that (i) both are white (ii) both are black.

(iii) one is black and other white

Soln:- (Total no. of ways of drawing ball from each bag $= {}^6C_1 \cdot {}^8C_1$)

Prob of getting white ball from first bag $= \frac{4}{6} = \frac{2}{3} = P(A_1)$ (say)

" " black ball " " " $= \frac{2}{6} = \frac{1}{3} = P(A_2)$

Prob of getting white ball from 2nd bag $= \frac{3}{8} = P(A_3)$

" black " " $= \frac{5}{8} = P(A_4)$

i) $P(\text{both white}) = P(A_1 \cap A_3)$

$= \frac{2}{3} \times \frac{3}{8} = \frac{2}{8}$

ii) $P(\text{both black}) = P(A_2 \cap A_4) = \frac{1}{3} \times \frac{5}{8} = \frac{5}{24}$

iii) $P(\text{one black \& other white}) = P(A_1 \cap A_4) + P(A_2 \cap A_3)$

$= \frac{2}{3} \times \frac{5}{8} + \frac{1}{3} \times \frac{3}{8} = \frac{10}{24} + \frac{3}{24} = \frac{13}{24}$

Probability associated with set theory:

Laws on set operations

i) $A \cap B = B \cap A$, $A \cup B = B \cup A$

ii) $A \cap (B \cap C) = (A \cap B) \cap C$; $A \cup (B \cup C) = (A \cup B) \cup C$

iii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

4) $\overline{A \cup B} = \bar{A} \cap \bar{B}$; $\overline{A \cap B} = \bar{A} \cup \bar{B}$

5) $\bar{\bar{A}} = A$

6) $A - B = A \cap \bar{B}$

If E_1 and E_2 are 2 events then,

① $E_1 \cup E_2$ is the event E_1 or E_2 or both.

② $E_1 \cap E_2$ is the event E_1 and also E_2.

③ E_1' is the event that occurs if E_1 does not occur.

④ $E_1 - E_2$ is the event E_1 but not E_2.

- The experiments when performed repeatedly given diff. results are called random experiments.
- A set S consisting of all possible outcomes of a random expt is called a sample space.
- An event E is a subset of the sample space S.
- A particular outcome that is an element in S is called a sample.

Axioms of probability

If S is a sample space, and E is the set of all events, then to each event A in E we associate a unique real number P(A). known as probability of event A. if the following axioms are satisfied.

① $P(S) = 1$

② For every event A in E. (probability of A) $0 \leq P(A) \leq 1$.

③ If $A_1, A_2, A_3, \ldots A_n$ are mutually exclusive events in E, then

$$P(A_1 \cup A_2 \cup A_3 \ldots \cup A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

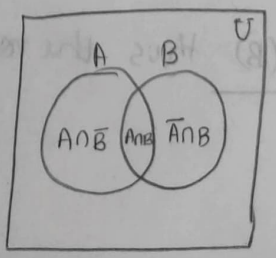
Addition rule:

If A and B are any 2 events in S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: $A = (A \cap \bar{B}) \cup (A \cap B)$

$B = (A \cap B) \cup (\bar{A} \cap B)$



$P(A) = P[(A \cap \bar{B}) \cup (A \cap B)] = P(A \cap \bar{B}) + P(A \cap B)$ by the 3rd axiom, since

$P(B) = P[(A \cap B) \cup (\bar{A} \cap B)] = P(A \cap B) + P(\bar{A} \cap B)$ $A \cap \bar{B}$ and $A \cap B$ are mutually exclusive

$P(A) + P(B) = P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(A \cap B) + P(\bar{A} \cap B)$

$= P(A \cup B) + P(A \cap B)$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$ — ①

If $A \cap B = \phi$
$P(A \cap B) = 0$
$\therefore P(A \cup B) = P(A) + P(B)$

Note:

If A and B are disjoint (mutually exclusive), we have $A \cap B = \phi$

$\therefore P(A \cap B) = 0$. Hence ① $\Rightarrow P(A \cup B) = P(A) + P(B)$.

Conditional probability

Let A and B be 2 events. Probability of happening of event B when the event A has already happened is called the conditional probability denoted by $P(B/A)$.

i.e, $P(B/A)$ is prob of B given A.

$$P(B/A) = \frac{\text{Prob of the happening of both events A \& B}}{\text{Prob of the happening of event A}} = \frac{P(A \cap B)}{P(A)}$$

$$\therefore P(B/A) = \frac{P(A \cap B)}{P(A)} \quad \text{— ①}$$

Why $P(A/B) = \frac{P(A \cap B)}{P(B)}$

Multiplication rule or product rule:

If A and B are any 2 events of S, then

$P(A \cap B) = P(A) \cdot P(B/A)$

Note: If A and B are independent events of S, then

$P(B/A) = P(B)$ thus the result becomes $P(A \cap B) = P(A) \cdot P(B)$

3) If A, B and C are ind. events, show that the following events are also independent.

i) $\bar{A}$ and $\bar{B}$ ii) $\bar{A}$ and B iii) A and $\bar{B}$ iv) $(A \cup B)$ and C

since A and B are independent, we have

$$P(A \cap B) = P(A) \cdot P(B) \quad \text{——— (1)}$$

i) W.K.T. $\overline{A \cup B} = \bar{A} \cap \bar{B}$ by De-Morgan's law.

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$$
$$= 1 - P(A \cup B)$$
$$= 1 - [P(A) + P(B) - P(A \cap B)]$$
$$= 1 - P(A) - P(B) + P(A) \cdot P(B) \quad \text{... from (2).}$$
$$= [1 - P(A)] - P(B)[1 - P(A)]$$
$$= [1 - P(A)][1 - P(B)]$$
$$= P(\bar{A}) \cdot P(\bar{B})$$

$\therefore \bar{A}$ and $\bar{B}$ are also independent events.

ii) $P(\bar{A} \cap B)$ $\quad (B - A = \bar{A} \cap B)$

We have $B = (A \cap B) \cup (\bar{A} \cap B)$

$$P(B) = P[(A \cap B) \cup (\bar{A} \cap B)]$$

$P(B) = P(A \cap B) + P(\bar{A} \cap B)$, since $(A \cap B)$ and $(\bar{A} \cap B)$ are disjoint.

$P(B) = P(A) \cdot P(B) + P(\bar{A} \cap B)$ from (2),

$$P(\bar{A} \cap B) = P(B) - P(A) \cdot P(B)$$
$$= P(B)[1 - P(A)]$$

$\therefore P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B)$

$\therefore \bar{A}$ and B are ind. events.

iii) A and $\bar{B}$

W.k.t $A = (A \cap B) \cup (A \cap \bar{B})$

$P(A) = P[(A \cap B) \cup (A \cap \bar{B})]$

$= P(A \cap B) + P(A \cap \bar{B})$

$= P(A) \cdot P(B) + P(A \cap \bar{B})$

$\therefore P(A \cap \bar{B}) = P(A) - P(A) \cdot P(B) = P(A)[1 - P(B)] = P(A) \cdot P(\bar{B})$

$\therefore$ A and $\bar{B}$ are independent events.

iv) $(A \cup B)$ and C.

Since A, B and C are independent events, so

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

so $P[(A \cup B) \cap C] = P[(A \cap C) \cup (B \cap C)]$ .. by distributive law

$= P(A \cap C) + P(B \cap C) - P[(A \cap C) \cap (B \cap C)]$ not disjoint

$= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$

$= P(A) \cdot P(C) + P(B) \cdot P(C) - \cancel{P[A \cap B \cap C]}$ $P(A) \cdot P(B) \cdot P(C)$

$P[(A \cup B) \cap C] = P(C)[P(A) + P(B) - P(A) \cdot P(B)]$

$= P(C)[P(A) + P(B) - P(A \cap B)]$. from ②

$= P(C) \cdot P(A \cup B)$

$\therefore (A \cup B)$ and C are independent events

2) If A and B are events with $P(A \cup B) = \frac{7}{8}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{5}{8}$. Find $P(A)$, $P(B)$ and $P(A \cap \bar{B})$.

soln:- W.k.t $P(\bar{A}) = \frac{5}{8}$

$P(A) = 1 - P(\bar{A}) = 1 - \frac{5}{8} = \frac{3}{8}$

W.k.t $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{7}{8} = \frac{3}{8} + P(B) - \frac{1}{4}$ $\Rightarrow \frac{7}{8} = \frac{1}{8} + P(B)$

$\therefore P(B) = \frac{6}{8} = \frac{3}{4}$

$A = (A \cap B) \cup (A \cap \bar{B})$

$P(A) = P[(A \cap B) \cup (A \cap \bar{B})]$

$P(A \cap \bar{B}) = P(A) - P(A \cap B)$

$= \frac{3}{8} - \frac{1}{4} = \frac{1}{8}$ //

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1) If A and B are events with $P(A) = \frac{1}{2}$, $P(A \cup B) = \frac{3}{4}$ & $P(\bar{B}) = \frac{5}{8}$.

Find $P(A \cap B)$, $P(\bar{A} \cap \bar{B})$, $P(\bar{A} \cup \bar{B})$ & $P(\bar{A} \cap B)$.

Given: $P(\bar{B}) = \frac{5}{8}$ $\therefore P(B) = \frac{3}{8}$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

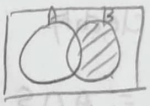
$\frac{3}{4} = \frac{1 \times 4}{2 \times 4} + \frac{3}{8} - P(A \cap B)$

$P(A \cap B) = \frac{7}{8} - \frac{3}{4} = \frac{1}{8}$.

$\bar{A} \cap \bar{B} = \overline{A \cup B}$ (Demorgan's law)

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{3}{4} = \frac{1}{4}$

$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - \frac{1}{8} = \frac{7}{8}$



$P(\bar{A} \cap B) = P(B) - P(A \cap B)$

$= \frac{3}{8} - \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$

2) P.T. $P(A \cup B \cup C) = P(A) + P(B) + P(C) + P(A \cap B \cap C) - P(A \cap B) - P(B \cap C) - P(A \cap C)$

Proof:

Let $A \cup B \cup C = A \cup (B \cup C)$

$P(A \cup B \cup C) = P(A \cup (B \cup C))$

$= P(A) + P(B \cup C) - P(A \cap (B \cup C))$

$= P(A) + P(B \cup C) + [P(B) + P(C) - P(B \cap C)] - P[(A \cap B) \cup (A \cap C)]$

$= P(A) + P(B) + P(C) - P(B \cap C) - [P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C))]$

$\therefore P(A \cup B \cup C) = P(A) + P(B) + P(C) + P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$.

3) Compute P(A) and P(B), given $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ & $P(\bar{B}) = \frac{1}{2}$.

Hence S.T. A and B are independent.

$P(\bar{B}) = \frac{1}{2} \Rightarrow P(B) = \frac{1}{2}$

Wkt $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{5}{6} = P(A) + \frac{1}{2} - \frac{1}{3}$

$\frac{5}{6} = \frac{1}{6} + P(A)$

$\therefore P(A) = \frac{4}{6} = \frac{2}{3}$

$P(A) \cdot P(B) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$ $\quad \therefore P(A \cap B) = P(A) \cdot P(B)$

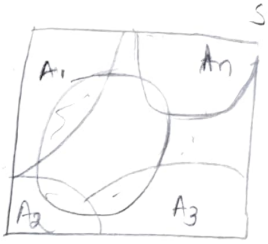
also $P(A \cap B) = \frac{1}{3}$ $\quad \therefore A$ & B are independent

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Baye's theorem :

Statement : Let $A_1, A_2, A_3, \ldots A_n$ be 'n' number of exhaustive, mutually exclusive events of the sample space 'S' with $P(A) \neq 0$ for each i. Let A be any other event associated with A_i's with $P(A) \neq 0$. Then

$$P(A_i/A) = \frac{P(A_i)\,P(A/A_i)}{\sum_{i=1}^{n} P(A_i)\,P(A/A_i)}$$



Proof: clearly $A \subset S$ & $S = A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_n$

$A = A \cap S$

$= A \cap (A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_n)$

$A = (A \cap A_1) \cup (A \cap A_2) \cup (A \cap A_3) \ldots \cup (A \cap A_n)$ ..(By distributive law)

$P(A) = P[(A \cap A_1) \cup (A \cap A_2) \cup (A \cap A_3) \ldots \cup (A \cap A_n)]$

$= P(A \cap A_1) + P(A \cap A_2) + P(A \cap A_3) + \ldots + P(A \cap A_n)$ by a special case of addition rule & the events $A \cap A_1, A \cap A_2, \ldots A \cap A_n$ are mutually exclusive events

$P(A/B) = \frac{P(A \cap B)}{P(B)}$

$P(A \cap B) = P(B) \cdot P(A/B)$

$P(A) = P(A_1)\,P(A/A_1) + P(A_2)\,P(A/A_2) + P(A_3)\,P(A/A_3) + \ldots P(A_n)\,P(A/A_n)$

$P(A) = \sum_{i=1}^{n} P(A_i)\,P(A/A_i)$ ——①

consider $P(A_i/A) = \frac{P(A \cap A_i)}{P(A)}$

$P(A_i/A) = \frac{P(A_i)\,P(A/A_i)}{\sum_{i=1}^{n} P(A_i)\,P(A/A_i)}$

1) 3 machines A, B and C produces 50%, 30% and 20% of the items of a factory. The percentage of defective outputs are 3, 4 and 5. If an item is selected at random, what is the prob that it is defective? What is the prob that it is from A?

Soln:- ~~P(A) = 0.5 P(B) = 0.3 P(C) = 0.2~~

Let D : (Probability) be the event of getting defective output.

Let X, Y and Z be the events of getting item produced by machine A, B and C resp.

$P(X) = \frac{50}{100} = 0.5$ $P(Y) = 0.3$ $P(Z) = 0.2$

$P(D/X) = 3\% = \frac{3}{100} = 0.03$ $P(D/Y) = 0.04$ $P(D/Z) = 0.05$

The probability of getting defective item is

$$P(D) = P(X)P(D/X) + P(Y)P(D/Y) + P(Z)P(D/Z)$$
$$= (0.5)(0.03) + (0.3)(0.04) + (0.2)(0.05)$$
$$= 0.015 + 0.012 + 0.01$$
$$= \underline{0.037}$$

The probability of getting an item produced by machine A, which is found defective is, $P(X/D) = \frac{P(X)P(D/X)}{P(D)} = \frac{0.5 \times 0.03}{0.037} = \underline{0.405}$

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2) The chance that a doctor will diagnose a disease correctly is 60%. The chance that a patient will die after diagnose is 40% and the chance of death by wrong diagnose is 70%. If a patient dies, what is the chance that this disease was correctly diagnosed?

Soln:- ~~Let A & B be 2 events that a patient~~

A : event of correct diagnosis $P(A) = \frac{60}{100} = 0.6$

B : " , wrong diagnosis $P(B) = 0.4$

D : event of patient death. $P(D/A) = 40\% = 0.4$ & $P(D/B) = 0.7$

$$P(A/D) = \frac{P(A)P(D/A)}{P(A)P(D/A) + P(B)P(D/B)}$$
$$= \frac{0.6 \times 0.4}{0.6 \times 0.4 + 0.4 \times 0.7} = \frac{0.24}{0.24 + 0.28} = \frac{0.24}{0.52} = \underline{0.461}$$

3) In a class, 70% of boys and 30% of girls. 5% of boys & 3% of girls are irregular to classes. What is the prob. of a student selected at random is irregular to the classes? and what is the prob that the irregular student is a girl?

Soln: Let B : selecting of a boy from class ; $P(B) = \frac{70}{100} = 0.7$

G : " girl " ; $P(G) = 0.3$

I : selecting an irregular student to the class.

$P(I/B) = 5\% = 0.05$ and $P(I/G) = 0.03$

Probability of a student selected at random is irregular to the classes.

$\therefore P(I) = P(B)\,P(I/B) + P(G)\,P(I/G)$

$= (0.7)(0.05) + (0.3)(0.03) = 0.035 + 0.009 = 0.044$ ~~0.125~~

Probability that the irregular student is a girl,

$P(G/I) = \frac{P(G)\,P(I/G)}{P(I)} = \frac{(0.3)(0.03)}{0.044} = \frac{0.009}{0.044} = 0.204$ ~~0.072 = 0.08~~

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4) 3 major parties A, B and C are contending for power in the elections of the state. and the chance of their winning in the elect is in the ratio 1:3:5. The parties A, B and C resp. have probabilities of banning the online lottery is $\frac{2}{3}, \frac{1}{3}, \frac{3}{5}$. What is the prob that there will be ban on online lottery in the state? What is the prob that the ban is from party C?

Soln: A : chance of winning election by party A

B : " " " B

C : " " " C

D : Event of banning the online Lottery.

$P(A) = \frac{1}{9}$ $P(B) = \frac{3}{9} = \frac{1}{3}$ $P(C) = \frac{5}{9}$

$P(D/A) = \frac{2}{3}$ $P(D/B) = \frac{1}{3}$ $P(D/C) = \frac{3}{5}$

Probability of banning online lottery,

$$P(D) = P(A)\,P(D/A) + P(B)\,P(D/B) + P(C)\,P(D/C)$$

$$= \frac{1}{9} \times \frac{2}{3} + \frac{1}{3} \times \frac{1}{3} + \frac{5}{9} \times \frac{3}{5}$$

$$= \frac{2}{27} + \frac{1}{9} + \frac{15}{45}$$

$$= 0.074 + 0.112 + 0.334$$

$$= \underline{0.52} \qquad 0.5185$$

Prob that the ban is from party C,

$$P(C/D) = \frac{P(C)\,P(D/C)}{P(D)} = \frac{\frac{5}{9} \times \frac{3}{5}}{0.52} = \underline{0.634} \qquad 0.6429$$

---

5) A bag contains 3 coins one of which is 2-headed and the other two normal and fair. A coin is chosen at random from the bag and tossed 4 times in succession. If head turns up each time, what is the prob that this is 2-headed coin?

Soln:- C_3 : choosing 2 headed coin

C_1 : } 2 normal coins.
C_2 :

chance of getting 2-headed coin, $P(C_3) = \frac{1}{3} = P(C_1) = P(C_2)$

H: event of getting head all 4 times.

$$P(H/C_1) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16} = P(H/C_2)$$

$$P(H/C_3) = \frac{2}{2} \times 1 \times 1 \times 1 = \underline{1}$$

$$\therefore P(C_3/H) = \frac{P(C_3)\,P(H/C_3)}{P(C_1)\,P(H/C_1) + P(C_2)\,P(H/C_2) + P(C_3)\,P(H/C_3)}$$

$$= \frac{\frac{1}{3} \times 1}{\left(\frac{1}{16}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{16}\right) + \left(\frac{1}{3}\right)(1)} = \frac{\frac{1}{3}}{\frac{18}{16 \times 3}} = \frac{16}{18} = \underline{0.889}$$

---

6) From 6 positive and 8 negative numbers, 4 no.s are choosen at random (without replacement) and multiplied. What is the probability that product is a +ve number?

Soln:-

Let P be a event of getting a +ve no.s

N " " " -ve no.s

To get the product of 4 no.s positive, we have the following 3 cases

(4P + 0N), (2P+2N), (0P + 4N)

The required probability ~~(for (4P+0N))~~ is $= \frac{{}^6C_4 \times {}^8C_0}{{}^{14}C_4} + \frac{{}^6C_2 \times {}^8C_2}{{}^{14}C_4} + \frac{{}^6C_0 \times {}^8C_4}{{}^{14}C_4}$

$= \frac{15 + 15 \times 28 + 70}{1001}$

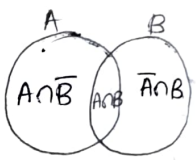
$= \frac{505}{1001} = 0.5045$

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7) If A and B are two events in S, s.t.

i) $P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A) + P(B) - 2P(A \cap B)$

ii) $P(\bar{A} \cap \bar{B}) = 1 + P(A \cap B) - P(A) - P(B)$



Here: $A = (A \cap \bar{B}) \cup (A \cap B)$

and $B = (\bar{A} \cap B) \cup (A \cap B)$

mutually exclusive

So, $P(A) = P[(A \cap \bar{B}) \cup (A \cap B)]$ & $P(B) = P[(\bar{A} \cap B) \cup (A \cap B)]$

Since $A \cap \bar{B}$ and $A \cap B$ are mutually exclusive, by the 3rd axiom of probability, we have

$P(A) = P(A \cap \bar{B}) + P(A \cap B)$ —①

Wly $P(B) = P(\bar{A} \cap B) + P(A \cap B)$ —②

① + ② ⇒ $P(A) + P(B) = P(A \cap \bar{B}) + P(\bar{A} \cap B) + 2P(A \cap B)$

so $P(A \cap \bar{B}) + P(\bar{A} \cap B) = P(A) + P(B) - 2P(A \cap B)$

Here $A \cap \bar{B}$ and $\bar{A} \cap B$ are again mutually exclusive

$\therefore P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A) + P(B) - 2P(A \cap B)$

ii) we have $\overline{A} \cap \overline{B} = \overline{A \cup B}$

so $P(\overline{A \cup B}) = 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$ Addition rule

$\therefore P(\overline{A} \cap \overline{B}) = 1 + P(A \cap B) - P(A) - P(B)$

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8) Given $P(A) = \frac{3}{4}$, $P(B) = \frac{1}{5}$ and $P(A \cap B) = \frac{1}{20}$. Find $P(A \cup B)$, $P(A \cap \overline{B})$, $P(\overline{A} \cap B)$, $P(A/B)$, $P(B/A)$, $P(\overline{A}/\overline{B})$, $P(\overline{B}/\overline{A})$, $P(A/\overline{B})$, $P(B/\overline{A})$.

Soln:- $P(A) = \frac{3}{4}$ $P(B) = \frac{1}{5}$ and $P(A \cap B) = \frac{1}{20}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{3}{4} + \frac{1}{5} - \frac{1}{20} = \frac{15+4}{20} - \frac{1}{20} = \frac{18}{20} = \frac{9}{10}$

$P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$= \frac{3}{4} - \frac{1}{20} = \frac{14}{20} = \frac{7}{10}$

$P(\overline{A} \cap B) = P(B) - P(A \cap B)$

$= \frac{1}{5} - \frac{1}{20} = \frac{3}{20}$

$P(A/B) = \frac{P(A \cap B)}{P(B)}$

$= \frac{\frac{1}{20}}{\frac{1}{5}} = \frac{1}{4}$

$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1/20}{3/4} = \frac{1}{15}$

$P(\overline{A}/\overline{B}) = \frac{P(\overline{A} \cap \overline{B})}{P(\overline{B})} = \frac{P(\overline{A \cup B})}{1 - P(B)} = \frac{1 - P(A \cup B)}{1 - P(B)}$

$P(\overline{B}/\overline{A}) = \frac{P(\overline{B} \cap \overline{A})}{P(\overline{A})} = \frac{1 - P(A \cup B)}{1 - P(A)} = \left(\frac{\frac{9}{10}}{\frac{1}{4}} = \frac{9}{10} \times 4 = \frac{18}{5}\right)$

$= \frac{1 - \frac{9}{10}}{1 - \frac{3}{4}} = \frac{\frac{1}{10}}{\frac{1}{3}} = \frac{3}{10}$

$P(A/\overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})} = \frac{\frac{7}{10}}{\frac{4}{5}} = \frac{7}{8}$

$P(B/\overline{A}) = \frac{P(B \cap \overline{A})}{P(\overline{A})} = \frac{\frac{3}{20}}{\frac{1}{4}} = \frac{3}{5}$

9) The prob that a team wins a match is 3/5. If this team plays 3 matches in a tournament. What is the prob that the team

i) win all the matches. $\frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$ iii) win almost one match

ii) win atleast one match $= \frac{27}{125}$ iv) lose all the matches.

10) In a bolt factory, there are 4 machines A, B, C and D manufacturing respectively 20%, 15%, 25% and 40% of the total production. Of these, 5%, 4%, 3% and 2% are defective. If a bolt is drawn at random is found defective, what is the prob that it was manufactured by either A or B ?

$$[P[(A \cup B)/X] = P(A/X) + P(B/X)]$$

Soln:-

9)

$$P(W_1) = P(W_2) = P(W_3) = \frac{3}{5}$$

i) $P(W_1 \cap W_2 \cap W_3) = \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$

$$= \frac{27}{125}$$

(ii) P(atleast 1) = 1 − P(losing all matches)

$$= 1 - P(\bar{W}_1)\, P(\bar{W}_2)\, P(\bar{W}_3)$$

$$= 1 - \left(\frac{2}{5}\right)\left(\frac{2}{5}\right)\left(\frac{2}{5}\right)$$

$$= 1 - \frac{8}{125} = \frac{117}{125}$$